JEFFREY HAMLIN VINOD

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PROFESSIONAL SUMMARY

Graduate Software Engineer pursuing an MSc in Computer Science (Data Science) at Trinity College Dublin, on track for First Class Honours. Hands-on experience designing and building scalable backend systems, RESTful APIs and distributed data pipelines with consistent performance under load. Proven ability to engineer and deploy ML-integrated applications across constrained and cloud-adjacent environments. Strong foundation in data structures, algorithms, operating systems and systems design with a data-driven approach to engineering decisions. Seeking an entry-level Software Engineering role contributing to large-scale, distributed, cloud-scale services.

EDUCATION

Trinity College Dublin: MSc Computer Science - Data Science | Expected First Class Honours (1:1) **Sep 2025 - Present**

Rajalakshmi Engineering College, India: B.E. Computer Science and Engineering | CGPA 8.01 / 10 **Oct 2021 - May 2025**

EXPERIENCE

Freelance Web Developer – Ananda Stores, Madurai (Link) ***Feb 2026 - Present***

- **Designed and shipped a full-stack B2B e-commerce platform -** translated client requirements for a wholesale business into a live, bilingual (English/Tamil) product catalogue using Vanilla JS, HTML5/CSS3, deployed to Vercel with Edge caching for optimised Core Web Vitals.
- **Built a secure REST API and admin dashboard -** engineered a Node.js/Express backend with PostgreSQL, implementing JWT-based stateless authentication, bcrypt password hashing and role-based access control restricting destructive operations to admin accounts.
- **Delivered scalable image management and dynamic CRUD operations -** integrated Cloudinary via Multer for direct image uploads, built real-time inventory analytics and deployed the full stack across Vercel (frontend) and Railway (API + PostgreSQL).

Frontend Developer Intern - Yi Chennai (Link) ***Aug 2023 - Oct 2023***

- **Built and shipped a production frontend -** architected a responsive, multi-section web platform using HTML, CSS and Bootstrap supporting 500+ active users, delivered from initial requirements through to launch.
- **Collaborated across teams to meet engineering goals -** worked with a 3-member engineering team and cross-functional stakeholders to align on requirements, gather feedback, and ensure delivery met business objectives.
- **Drove iterative improvement through structured feedback -** presented regular progress demos to stakeholders and translated feedback into implementation changes, maintaining delivery momentum throughout the project lifecycle.

TECHNICAL SKILLS

- **Languages:** Python, JavaScript, Java, C, SQL, HTML, CSS
- **Backend & Databases:** Node.js, REST APIs, SQLite, PostgreSQL, Git, GitHub, Railway, Debugging
- **Machine Learning:** TensorFlow, TensorFlow Lite, scikit-learn, ML model development and deployment
- **Tools & Platforms:** Linux/Unix, UiPath Studio, Power Automate, UiPath Cloud, D3.js, Power BI, Cloudinary, Vercel
- **Courses:** Data Structures & Algorithms, Operating Systems, Object-Oriented Design, Cloud Computing, Scalable Computing, Distributed Systems

PROJECTS

Personal Finance Tracker: *Python · Node.js · SQLite · REST APIs · ML*

- **Designed scalable backend architecture -** built a Node.js service with RESTful API design, structured data persistence and validation workflows to ensure reliable, consistent transaction management across concurrent operations.
- **Engineered high-throughput data pipeline -** ingested, cleaned and transformed raw CSV bank statements; stress-tested end-to-end against 500-10,000 records with stable, predictable performance at scale.
- **Implemented ML classification with telemetry -** built a transaction categorisation model achieving ~85% accuracy across 7 classes, with comprehensive test coverage and logging.

AI-Resistant Scalable CAPTCHA System: *Python · TensorFlow Lite · Raspberry Pi · ML/DL*

- **Engineered distributed behavioural telemetry system -** captured and processed real-time keystroke dynamics and mouse movement signals to achieve 100% bot rejection while maintaining 100% human pass rate across all test scenarios.
- **Deployed production ML model on constrained hardware -** trained a CRNN model on 4,000+ synthetic images, optimised and converted to TensorFlow Lite and shipped to a Raspberry Pi runtime with strict memory and compute constraints.
- **Designed fault-tolerant scoring engine -** implemented rule-based classification logic with 5-attempt retry mechanism and anomaly detection, ensuring system reliability and graceful degradation under adversarial input conditions.

Recruitment Automation Platform: *UiPath · WhatsApp API · Email · Excel*

- **Built production-grade automation pipeline -** engineered an end-to-end system dispatching personalised communications to hundreds of recipients from a single trigger, with zero manual intervention post-deployment.
- **Delivered lasting operational impact -** system adopted by the team and still in active use across multiple recruitment cycles, demonstrating reliability, maintainability, and real-world engineering quality.

LEADERSHIP & VOLUNTEERING

- Yi YUVA REC (Chair Sports Vertical): Managed multi-team volunteers and delivered 50+ community programs
- Recharge 25 (Communication & Sponsorship Lead): Coordinated outreach for a 150+ member cultural fest team.
- Competitive Programming Mentor for junior placement preparation